

HAZARDOUS-AREA CCTV INFRASTRUCTURE

Designed for the field. Built for control.

COROMANDEL INTERNATIONAL | DAHEJ SEZ II



CAMERA



FLAMEPROOF FIELD ASSEMBLY

Solution design, supply and implementation by D-TECH

57 cameras | 20 FLP enclosures | Fiber backbone | Central recording

A plant-wide CCTV platform engineered for scale.

The documented order combines surveillance, hazardous-area protection, fiber networking, recording, storage and control-room visualization for the Dahej site.

57

CAMERA ENDPOINTS

54 fixed bullet cameras and 3 PTZ cameras.

20

FLP ENCLOSURES

Outdoor bullet-camera housings for selected hazardous-area points.

192 TB

RAW DRIVE CAPACITY

32 surveillance hard disks, each 6 TB.

8.0 KM

ARMoured CABLING

6.5 km CAT6 plus 1.5 km six-core optical fiber.

THE OUTCOME

A multi-layer CCTV environment connecting protected field cameras, PoE access switching, fiber uplinks, centralized NVR recording, surveillance storage and security displays.

ORDER EVIDENCE

Purchase Order 130008242 dated 20-09-2022 lists 26 material categories. Commercial values and full order pages are excluded from this case study.

D-TECH VALUE

One accountable system integrator for field design, network architecture, recording, racks and deployment coordination.

Industrial surveillance must survive the environment.

A chemical plant requires more than camera supply: field protection, distance-aware connectivity, centralized recording and maintainable installation standards.

HAZARDOUS FIELD POINTS

Selected camera locations required flameproof outdoor housings and disciplined cable entry practices.

LONG-DISTANCE CONNECTIVITY

Distributed plant areas required a hybrid armored CAT6 and single-mode fiber backbone.

MIXED VIEW REQUIREMENTS

Fixed 2 MP, 4 MP color and PTZ cameras were needed for different operational scenes.

STORAGE AT SCALE

A 57-camera estate required multiple recorders and high-capacity surveillance storage.

OUTDOOR MOUNTING

Metal stands, 20-foot poles, junction boxes and power cabins were needed for stable field installation.

CONTROL-ROOM VISIBILITY

Large displays, HDMI links and rack infrastructure were needed for centralized monitoring.

DESIGN QUESTION

How can field cameras, hazardous-area protection, copper, fiber, PoE, recording and monitoring operate as one reliable plant surveillance platform?

Six engineering layers, one operational platform.

D-TECH structured the solution from field observation points through network transport and centralized monitoring.



DESIGN PRINCIPLE

Every camera point is treated as part of a documented field-to-control-room chain.

Field-to-control-room architecture.

The architecture separates hazardous-area field hardware from safe-zone network, recording and monitoring equipment.



SEGMENTATION & CONTROL

Camera addressing, switch uplinks, recorder allocation and user permissions should be documented so fault isolation and future expansion remain controlled.

RESILIENCE PATH

Spare fiber cores, rack capacity and distributed switching provide practical headroom for route changes, expansion and service recovery.

Camera and hazardous-area field equipment.

Conceptual line drawings show the function of primary field products. Final dimensions and approvals follow OEM drawings and plant engineering standards.



2 MP BULLET CAMERA

27 nos | Fixed surveillance



4 MP COLOR BULLET

27 nos | Higher-detail color view



PTZ CAMERA

3 nos | Active pan-tilt-zoom



FLP ENCLOSURE

20 nos | Outdoor bullet housing



METAL CAMERA STAND

34 nos | Field mounting support



20-FOOT POLE

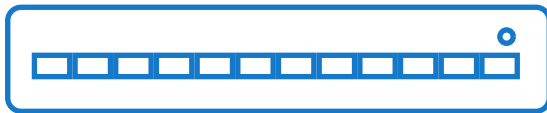
11 nos | Elevated observation point

FIELD ASSEMBLY

Camera + enclosure or outdoor housing + stand or pole + junction box + protected power/data route.

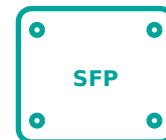
Network, fiber and distribution infrastructure.

The edge and backbone components create the transport layer between distributed camera points and centralized recording.



24-PORT PoE SWITCH

10 nos | 2 SFP uplinks



SFP OPTICAL MODULE

24 nos | Single-mode LC



CAT6 ARMOURED

ARMOURED CAT6

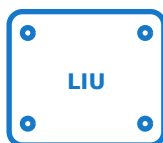
6,500 m | Camera edge cabling



6-CORE SINGLE MODE

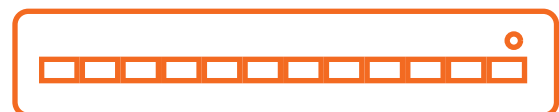
6-CORE ARMOURED OFC

1,500 m | Fiber backbone



12-PORT LOADED LIU

12 nos | Fiber termination



24-PORT PATCH PANEL

3 nos | Structured copper

PASSIVE SCOPE

Also includes SC-LC patch cords, CAT6 patch cords, RJ45 connectors and flexible braided PVC protection.

Recording, storage and control-room infrastructure.

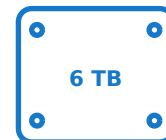
The centralized layer combines recorder capacity, surveillance drives, racks and large-format monitoring displays.



64-CHANNEL NVR

64-CHANNEL NVR

4 nos | DS-9664I-I8 reference



6 TB

6 TB SURVEILLANCE HDD

32 nos | 192 TB raw capacity



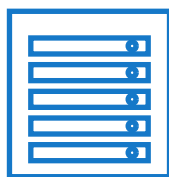
65-INCH SMART TV

2 nos | Security display



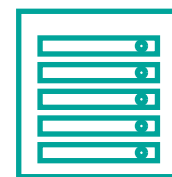
**42U / SYSTEM RACK
42U NETWORK RACK**

1 no | Core equipment rack



**15U / SYSTEM RACK
15U RACK**

1 no | Local equipment rack



**6U / SYSTEM RACK
6U WALL-MOUNT RACK**

6 nos | Edge distribution

ADDITIONAL SCOPE

PTZ stands, outdoor PTZ junction boxes, HDMI cables and an outdoor power-point cabin complete the operating environment.

Survey, stage, deploy and commission.

The implementation sequence protects design intent while coordinating civil, electrical, network and security stakeholders.



IMPLEMENTATION CONTROL

Work-area permits, shutdown coordination and approved hazardous-area practices govern execution at every field point.

Protection depends on every interface.

The enclosure alone does not create a compliant field point; classification, entry hardware, earthing, mounting and maintenance practice must work together.

ZONE VALIDATION

Confirm the plant-approved area classification, gas group and temperature class before final equipment acceptance.

ENCLOSURE SUITABILITY

Validate the supplied FLP housing model against the approved hazardous-area requirement and certification documentation.

CABLE ENTRY

Use approved glands, sealing methods, conduit transitions and bend radius at every protected field assembly.

EARTHING & BONDING

Bond metallic enclosures, stands, poles and associated equipment according to the plant electrical standard.

WEATHER PROTECTION

Protect outdoor junction boxes, power cabins and connectors against water, dust and mechanical exposure.

SAFE-ZONE EQUIPMENT

Keep standard switches, NVRs, displays and racks in appropriate safe areas unless independently certified.

DOCUMENTATION

Maintain camera IDs, route schedules, rack elevations, fiber cores, IP plan and acceptance records.

MAINTENANCE CONTROL

Use permits, isolation and approved opening procedures before servicing hazardous-area assemblies.

ENGINEERING NOTE

This case study summarizes the documented supply and D-TECH solution architecture. Final hazardous-area compliance must follow approved certificates, site classification, statutory requirements, OEM instructions and Coromandel engineering approvals.

CASE STUDY CONCLUSION

One architecture. From hazardous field to control room.

57 CAMERA ESTATE

Fixed and PTZ coverage structured for varied plant observation needs.

HAZARDOUS-AREA READY

Twenty FLP enclosures support protected bullet-camera locations.

FIBER-BACKED NETWORK

Armored copper and single-mode fiber connect distributed plant zones.

CENTRALIZED RECORDING

Four 64-channel NVRs and 192 TB raw drive capacity support retention.

MAINTAINABLE INFRA

Racks, LIUs, patching, poles, stands and junction hardware complete the system.

D-TECH OWNERSHIP

Design, supply and implementation coordinated as one end-to-end solution.

D-TECH: INDUSTRIAL CCTV, NETWORK, FIBER, RECORDING AND SUPPORT